MIT 18.06
Linear Algebra

Problem Set 2

#1 Section 2.5 - Q24

$$U = \left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 = R_1 - aR_2} \left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 = R_2 - cR_3} \left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0 \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 = R_1 - (b-ac)R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & -b+ac \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$= U^{-1}$

#2 Section 2.5 - Q40

$$A = \left[\begin{array}{cccc|cccc} 1 & -a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\substack{R_1 = R_1 + aR_2 \\ R_2 = R_2 + bR_3 \\ R_3 = R_3 + cR_4}]{} \left[\begin{array}{cccc|cccc} 1 & 0 & -ab & 0 & 1 & a & 0 & 0 \\ 0 & 1 & 0 & -bc & 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\substack{R_1 = R_1 + abR_3 \\ R_2 = R_2 + bcR_4}]{} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & a & ab & abc \\ 0 & 1 & 0 & 0 & 0 & 1 & b & bc \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$= A^{-1}$

#3 Section 2.6 — Q13

$$A=\begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \xrightarrow[\substack{R_2=R_2-R_1 \\ R_3=R_3-R_1 \\ R_4=R_4-R_1}]{} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix} \qquad E=\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ -1 & -1 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow[\substack{R_3=R_3-R_2 \\ R_4=R_4-R_2}]{} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix} \xrightarrow[R_4=R_4-R_3]{} \underbrace{\begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}}_{=U} \qquad L=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

To get 4 pivots for A, $a\neq 0$, $b\neq a$, $c\neq b$, $d\neq c$

#4 Section 2.6 — Q18

$$LDU = L_1D_1U_1 \rightarrow L_1^{-1}LDU = L_1^{-1}L_1D_1U_1$$

$$\rightarrow L_1^{-1}LD\underbrace{UU^{-1}}_{I} = \underbrace{L_1^{-1}L_1}_{I}D_1U_1U^{-1} \rightarrow \underline{L_1^{-1}LD = D_1U_1U^{-1}}$$

$L_1^{-1}LD$ is lower triangular, $D_1U_1U^{-1}$ is upper triangular, they are equal, so they are both <u>diagonal</u>,

so $L_1^{-1}L$ and U_1U^{-1} are diagonal, they all have diagonal 1's,

so $L_1^{-1}L=I$, $L=L_1$, $U_1U^{-1}=I$, $U=U_1$, $L^{-1}LD=D_1UU^{-1}$, $D=D_1$

#5 Section 2.6 — Q25

See the file ps2.ipynb in problem_code folder

$L^{-1}[i][j] = \frac{j}{i}$ for $j\leq i$ / 0 for $j>i$

#6 Section 2.6 - Q26

$$7K^{-1} = \begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 1 \\ 5 & 10 & 8 & 6 & 4 & 2 \\ 4 & 8 & 12 & 9 & 6 & 3 \\ 3 & 6 & 9 & 12 & 8 & 4 \\ 2 & 4 & 6 & 8 & 10 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$$

See the file ps2.ipynb in problem_file folder.

$$K \cdot 7K^{-1} = 7I$$

#7 Section 2.7 - Q13

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = P_2$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = P_3 = I$$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad P^3 = I, \quad P^4 \neq I$$

#8 Section 2.7 - Q36

L: Yes, S: No, M: No, D: Yes, P: Yes, Q: Yes

1. all scalar products $c \cdot P$ where c is a scalar, P is a permutation matrix
2. the block matrix $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$

#9 Section 2.7 - Q40

(a) $Q^T = Q^{-1}$, $Q^T Q_{ij} = Q_i^T Q_j$, $Q^T Q = I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$

$\|q_i\|^2 = q_i^T q_i = 1$ in I

(b) $q_i^T q_j = 0$ in I

(c) $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

#10 Section 3.1 - Q 18

(a) $A^T = A$, $B^T = B$, $(A+B)^T = A^T + B^T = A + B$, true

(b) $A^T = -A$, $B^T = -B$, $(A+B)^T = A^T + B^T = -A - B = -(A+B)$, true

(c) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, false

#11 Section 3.1 - Q23

- b is a linear combination of columns of A
- get larger: $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$
- don't get large: $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$
- you need b to be contained in $C(A)$ such that $Ax = b$ has a solution.

#12 Section 2.7 - Question 30

(a) Say s_1, s_2 in S, t_1, t_2 in T,

$(s_1+t_1)+(s_2+t_2)=(s_1+s_2)+(t_1+t_2)$ in $S+T$

$c(s_1+t_1)=cs_1+ct_1$ in $S+T$

(b) $S+T$ is a plane,

the span $S \cup T$ is the set of all combinations of vectors in the union.

the span $S \cup T$ contains all $s+t$, all $s+t$ forms $S+T$

#13 Section 2.7 - Q 32

- each column in AB is a linear combination of columns in A, each col in $[A \ AB]$ is a linear combination of columns in A, so $[A \ AB]$ and A have the same column space.
- $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $C(A^2)=Z$,
- $C(A)=R^n$ exactly when A is an invertible matrix.